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MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA(Honours) REGULAREXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBTG301 - BASIC

BIOINFORMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Word Answer Questions-Answer any 10 questions

Each question carries 2 marks

1. State one importance of literature mining. **[K] / [CO1]**
   It helps identify research gaps and find scientific information- **2 marks**
2. What do you mean by a nucleotide sequence database? **[K] / [CO2]**
   A database that stores DNA and RNA sequences- **2 marks.**
3. What type of information is used to build a MODELLER model? **[U] / [CO3]**
   A template protein structure and sequence alignment- **2 marks.**
4. Name one type of phylogenetic tree based on branching pattern. **[U] / [CO3]**
   Rooted tree or unrooted tree.- **2 marks.**
5. What does RMSD measure in Homology Modelling. **[U] / [CO3]**
   It measures the difference between two structures.- **2 marks**.
6. Identify How does Map Viewer represent genomic information? **[U] / [CO3]**
   It shows chromosomes, genes and other genomic features- **2 marks**
7. State the main use of RasMol. **[U] / [CO3]**
   To view three-dimensional molecular structures.- **2 marks.**
8. Distinguish between SBDD and LBDD. **[U] / [CO4]**
   SBDD uses the target structure; LBDD uses information about known ligands.Or SBDD: Structure-Based Drug Design; LBDD: Ligand-Based Drug Design- **2 marks**.
9. What is confidentiality in bioinformatics? **[K] / [CO4]**
   Keeping personal and biological data safe from unauthorized access.- **2 marks**
10. Differentiate between a hydrogen-bond donor and a hydrogen-bond acceptor. **[U] / [CO4]**
    A donor provides hydrogen; an acceptor receives/interacts with it- **2 marks**.

11. What is a binding site? **[K] / [CO4]**

The region of a molecule where a ligand binds- **2 marks.**

12. Define the computational study of an organism's complete genetic material called?. called**?[K] / [CO4]**

Computational genomics or genome informatics- **2 marks**.

[2x10 = 20]

Part B

Short Essay Questions-Answer any 4 questions

Each question carries 5 marks

13. Explain how to retrieve gene information using Ensembl. **[A] / [CO2]**

Open the Ensembl website and choose the species→Enter the gene name or ID in the search box. Choose the correct gene from the results→Check the gene summary, location and structure→View or download sequence and other gene details.- **5 marks**

14. Explain the major uses of GenBank in bioinformatics. **[U] / [CO2]**

Stores DNA and RNA sequences→Gives each sequence a unique accession number→Provides information about genes and coding regions→Helps compare sequences→Supports data sharing and biological research- **5 marks**

15. Describe the basic principle of an ORF Viewer. **[U] / [CO3]**

An ORF is a possible protein-coding region, The tool identifies start codons. It identifies stop codons. It can check the six reading frames of DNA, It displays the positions and lengths of possible ORFs. - **5 marks**

16. Demonstrate how insertions, deletions and mutations can be detected using dot plots. **[S] / [CO3]**

A dot plot compares two sequences.- **1 mark**

A continuous diagonal shows sequence similarity-**1 mark**

An insertion may cause a shift in the diagonal-**1 mark**

A deletion may also cause a shift or break-**1 mark**

A substitution may interrupt or weaken the diagonal- **l mark.**

17. Explain the role of eDNA and geospatial analysis in biodiversity assessment. **[An] / [CO4]**

Environmental DNA (eDNA) is DNA collected from samples such as water or soil- **1 mark.**

Role-It helps detect species from environmental samples, It can identify species that are difficult to observe, GIS maps species locations and habitats, Combining eDNA and maps helps monitor biodiversity and plan conservation- **4 marks.**

18. Describe the application of bioinformatics in pharmacogenomics and predicting adverse drug reactions. **[An] / [CO4]**

Pharmacogenomics studies how genes affect drug response- **2marks.**

Genetic variations such as SNPs can affect drug metabolism. Bioinformatics tools help analyse these variations- **1 mark,** This information can help to predict side effects or poor drug response**- 1 mark.** It supports safer drug choice and suitable dosing- **1 mark.**

[5x4 = 20]

Part C

Essay Questions-Answer any 2 questions

Each question carries 15 marks

19. Describe the Human Genome Project and explain its objectives, methodology and major achievements. **[U] / [CO1]**

    Introduction and definition -**2 marks**, Objectives -**3 marks**, Methods **4 marks**, Achievements -**4 marks** Significance - **2 marks**

20. Describe the major primary sequence databases and explain their role in storing biological sequence information. **[U] / [CO2]**

    Definition -**2 marks,** GenBank -**2 marks**, ENA - EMBL-EBI -**2 marks**, DDBJ -**2 marks,** Data sharing -GenBank, ENA and DDBJ -**2 marks**, Records and annotations -**2 marks.** Uses -**3 marks.**

21. Compare the Needleman-Wunsch and Smith-Waterman algorithms with respect to principle, scoring and Applications **[An] / [CO3]**

    Basic principle -**2 marks**, Needleman–Wunsch **-3 marks**, Smith–Waterman **-3 marks**, Scoring -**3 marks**, Main differences -**2 marks**, Applications **-2 marks**.

22. Evaluate the concept of Classical Structure–Activity Relationship (SAR) and describe the major steps involved. **[An] / [CO4]**

    Definition -**2 marks** ,Structure and activity -**2 marks,** Select compounds -**2 marks**, Modify structures -**2 marks,** Test compounds -**2 marks**, Compare results -**3 marks** , Uses and limitations -**2 marks**

[15x2 = 30]